MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

PROJECTILE 5/54

IN PALLET ADAPTER MK 11 AND MODS

FLEET ISSUE UNIT LOAD

MIL-STD-1320-152A

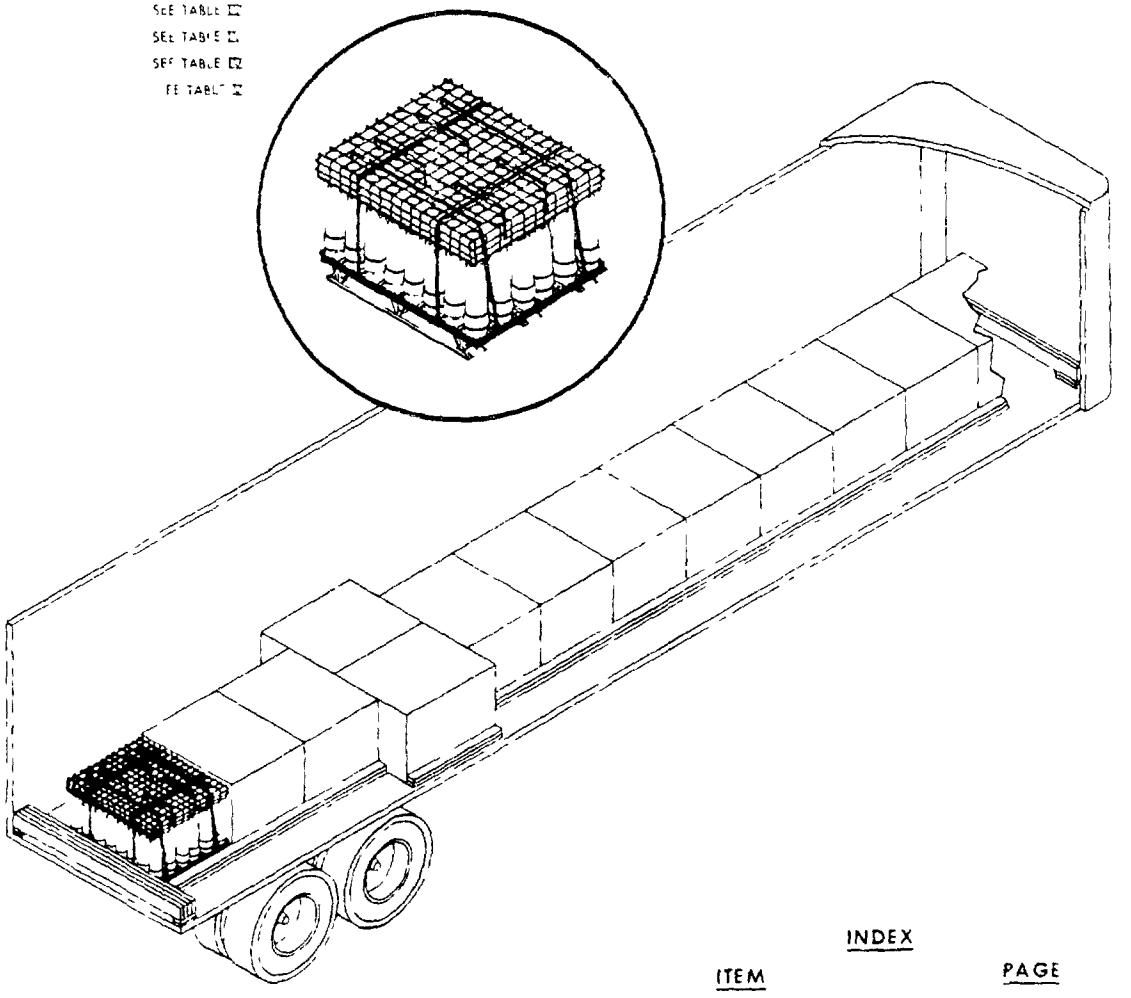
(NAVY)

30 SEPTEMBER 1975

SUPERSEDING
MIL STD 1320-152 (NAVY)
10 SEPTEMBER 1974

UNIT LOAD DATA

UNIT LOAD DRAWING	MK [illegible] MOD [illegible]
DIMENSIONS	SEE TABLE IX
CUBE	SEE TABLE IX
WEIGHT	SEE TABLE IX
DOT HAZARD CLASS	SEE TABLE I



NOTE

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARD REFER TO INDEX TO STANDARDS (MIL-HDBK-236 (NAVY)

INDEX

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FTL	2 & 4 & 5
LTL	3 & 6

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M L Gray NWHL 6/11/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

T E Kelly 6/11/75
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR [signature]
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF 5"/54 PROJECTILES PALLETIZED IN IN ACCORDANCE WITH WR-54/6 AND WR-54/271

2 THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 42 44 AND 45 FT TRAILERS WITH WOOD FLOORS METAL FLOORS AND METAL FLOORS WITH WOOD NAILING STRIPS THE TRAILERS SHALL HAVE THE TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER)

NOTE

TO FIND THE CORRECT LOAD PATTERN TO BE USED

a. DETERMINE THE TYPE OF PROJECTILES TO BE LOADED (5"/54 PROJECTILES WITH WATERPROOF CAP 5"/54 PROJECTILES WITH PLASTIC ROTATING BANDS OR ROCKET ASSISTED PROJECTILES)

b. SELECT THE CORRECT TABLE FOR THE TYPE OF PROJECTILES BEING LOADED (TABLE I OR TABLE II)

c. DETERMINE THE LENGTH OF THE TRAILER BEING LOADED

d. FIND THE LOAD PATTERN NUMBER IN THE SELECTED TABLE USING THE NUMBER OF UNIT LOADS *BEING SHIPPED AND THE LENGTH OF THE TRAILER BEING LOADED*

e. USING THE LOAD PATTERN NUMBER ENTER TABLE III AND FIND THE LOAD PATTERN TO BE USED

f. LOAD BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 4 5 AND 6

3 THE CORRECT REAR BLOCKING TO BE USED DEPENDS UPON THE DISTANCE BETWEEN THE REAR OF THE LADING AND THE TRAILER DOORS WHEN CLOSED WHEN THAT DISTANCE IS

LESS THAN 12 INCHES

INSTALL REAR BLOCKING ASSEMBLY DETAIL B PAGE 4 AS SHOWN ON PAGE 5

CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

12 TO 36 INCHES

INSTALL REAR BLOCKING ASSEMBLY DETAIL C PAGE 4 IN LIEU OF REAR BLOCKING SHOWN ON PAGE 5

CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

GREATER THAN 36 INCHES

INSTALL REAR BLOCKING AS SHOWN FOR LTL ON PAGE 6 DO NOT USE TRAILERS WITH METAL FLOORS

4 THE BLOCKING METHODS SHOWN ON PAGES 5 AND 6 ARE FOR WOOD FLOOR TRAILERS WHEN THE TRAILER HAS A METAL FLOOR OR A METAL FLOOR WITH WOOD NAILING STRIPS AND SLEEPERS CAN NOT BE USED TO PREVENT LATERAL MOTION INSTALL FILLER ASSEMBLY, DETAIL D PAGE 4

5 FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE UNIT LOAD AND THE TRAILER SIDE WALLS THEY SHALL BE SECURED IN PLACE *TO PREVENT LONGITUDINAL MOVEMENT*

WARNING

THE USE OF FILLER ASSEMBLIES WHEN SHIPPING THE HEAVIER FTL'S (44 750 LBS & 45 448 LBS) MAY CAUSE THE VEHICLE TO BE OVERWEIGHT GROSS VEHICLE WEIGHTS AND AXLE WEIGHT SHOULD BE COMPUTED PRIOR TO LOADING

6 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

7 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO THE INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS ATTACH APPROPRIATE PLACARD TO BOTH SIDE FRONT AND REAR OF TRAILER AS SPECIFIED FOR THE TYPE OF PROJECTILE IN TABLE V, PAGE 4

8 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAILS FF-N-105 TYPE I STYLE IC COMMON BRIGHT

9 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY)

FTL 40, 42, 44 & 45 FT TRAILERS (VAN)

1 A FULL TRUCKLOAD CONSISTS OF

a. ELEVEN UNIT LOADS OF PROJECTILES WITH WATERPROOF CAPS OR TWELVE UNIT LOADS OF PROJECTILES WITH PLASTIC ROTATING BANDS INTO A STANDARD WEIGHT TRACTOR TRAILER (NET VEHICLE WEIGHT 27 500 LBS MAX.)

b. TWELVE UNIT LOADS OF PROJECTILES WITH WATERPROOF CAPS OR THIRTEEN UNIT LOADS OF PROJECTILES WITH PLASTIC ROTATING BANDS INTO A LIGHT WEIGHT TRACTOR TRAILER (NET VEHICLE WEIGHT 25 500 LBS MAX.)

2 WHEN LOADING 45 FT TRAILERS, USE THE SAME LOAD PLAN AND PROCEDURES AS SPECIFIED FOR THE 44 FT TRAILERS, INCREASING THE THICKNESS OF THE FRONT BULKHEAD BY SIX INCHES

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TABLE I

PROJECTILES WATERPROOF CAP & PLASTIC RATATING BAND

NO. OF UNIT LOADS TO BE SHIPPED	FTL OR LTL	LOAD PATTERN NO.		
		TRAILER LENGTH		
		40	47	44
2	FTL	4**	5**	6**
	FTL	7	8	7
	LTL	7	10	6
7	LTL	10	2	2
8	LTL		14	4
	LTL	7		10
NO. OF LEG	SEE TABLE III			

TABLE II

ROCKET ASSISTED PROJECTILE MK 58 MOD

NO. OF UNIT LOADS TO BE SHIPPED	FTL OR LTL	LOAD PATTERN NO.		
		TRAILER LENGTH		
		40	42	44
3	FTL	3*	2*	*
4	FTL	4	5	5
7	LTL	9	8	
10	LTL	7	10	6
9	LTL	12	12	7
8	TL	15	14	4
7	LTL	17	17	
NO OF LEG	SEE TABLE III			

* **WARNING**

THIS LOAD IS SUITABLE FOR LIGHTWEIGHT TRACTOR TRAILERS ONLY. IF LOADED ON STANDARD WEIGHT TRACTOR TRAILER, THE AXLE WEIGHT AND GROSS VEHICLE WEIGHT WILL BE IN EXCESS OF THE LIMITS OF 32,000 LB FOR TANDEM AXLE AND 73,280 LBS GROSS VEHICLE WEIGHT.

* **NOTE**

WHEN LOADING THESE UNIT LOADS OF HI-FRAG PROJECTILES STANDARD WEIGHT TRACTOR TRAILER MAY BE USED

TABLE III

LOAD PATTERN FWD — PATTERN NUMBER — LOAD PATTERN FWD — PATTERN NUMBER — LOAD PATTERN FWD

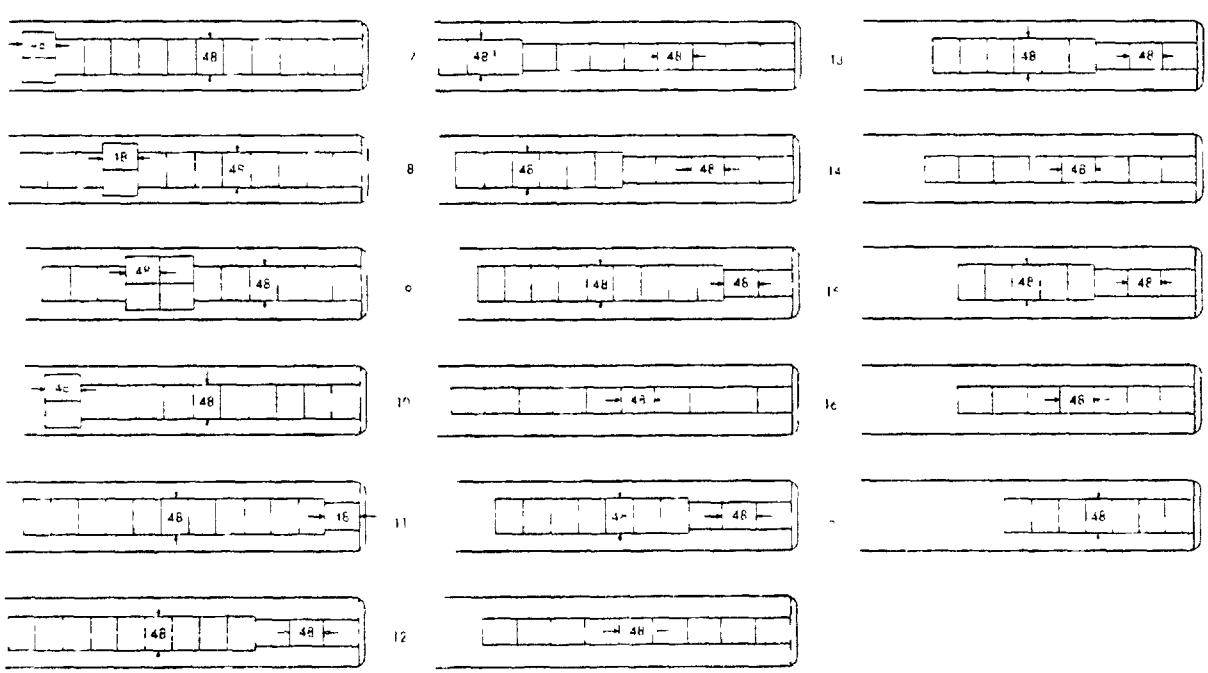

48
48
48

TABLE IV

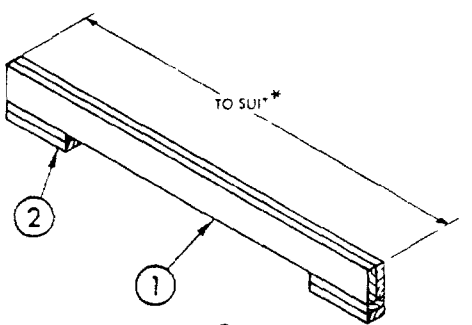
ITEM	DIMENSIONS L x W x H	CU FT	G L WT (LBS * ADAPTER MK 1 MOD 0	G L WT (LBS * ADAPTER MK 1 MOD 1	NO OF UNIT LOADS FTL	FTL WT (LBS * ADAPTER MK 1 MOD 0	FTL WT (LBS * ADAPTER MK 1 MOD 1
PROJECTILE WITH WATERPROOF CAP	40 x 48 x 33 3/4	37.5	377**	3 22	12	45 348	44 676
ROCKET ASSISTED PROJECTILE MI 58 MOD 0	40 x 48 x 34	37.8	3441	3387	1	44 759	44 031
PROJECTILE WITH PLASTIC ROTATING BAND (HI-FRAG	40 x 48 x 34 1/2	38.0	3637 **		12	43 584	

*WEIGHTS ARE APPROXIMATE

**PALLETIZED IN MK 11 MOD 2 BOTTOM AND MK 11 MOD 1 TOP

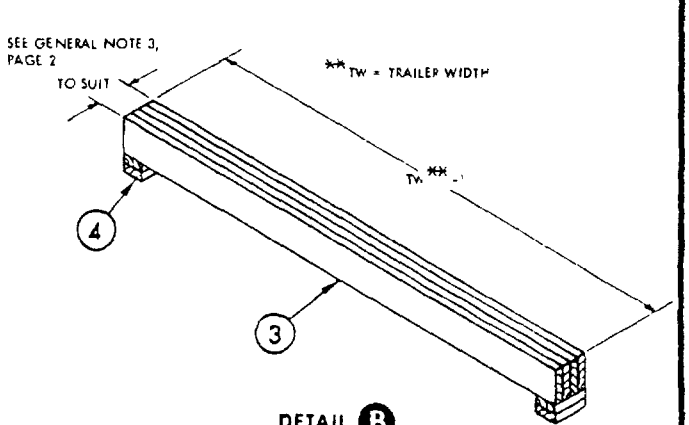
TABLE V

HAZARD CLASSIFICATION			
TYPE OF PROJECTILE	DOT	TYPE OF PROJECTILE	DOT
TP	B	COM	A
HE-CVT	A	IR	A
CHAFF	B	ILLUM	B
SMK-WP	A	HE-PD	A
VT	A	VT-NF	B
AAC	A	IR-NF	B
HC	A	HI-FRAC	A



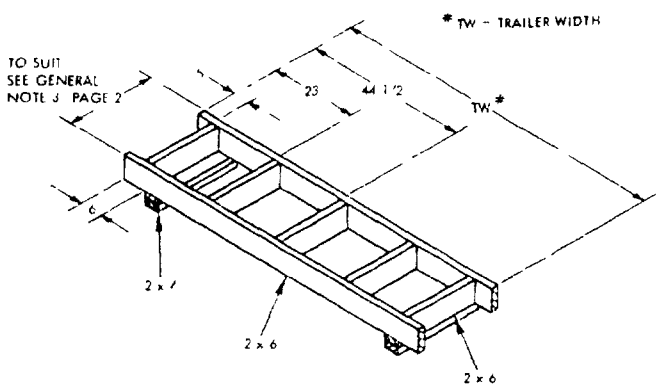
DETAIL A

FRONT BULKHEAD
1 REQUIRED



DETAIL B

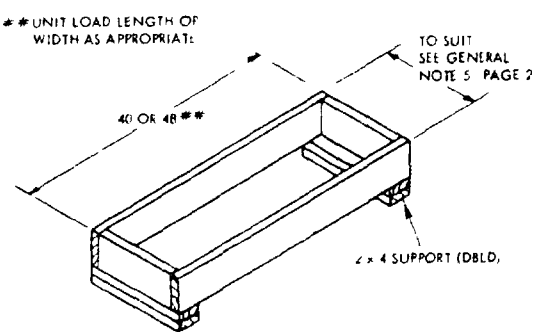
REAR BLOCKING ASSEMBLY
1 REQUIRED



DETAIL C

REAR BLOCKING ASSEMBLY

USE WHEN SPACE BETWEEN LADING AND DOORS IS 32 TO 36 INCHES (SEE GENERAL NOTE 3, PAGE 2

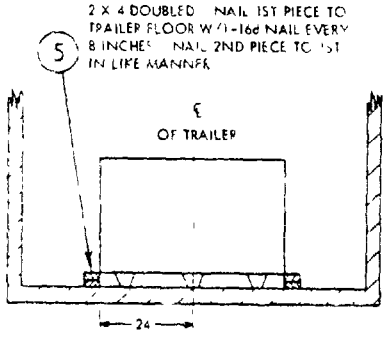


DETAIL D

FILLER ASSEMBLY

ALTERNATE, TO BE USED WHEN SLEEPERS CAN NOT BE USED SEE GENERAL NOTE 4 PAGE 2

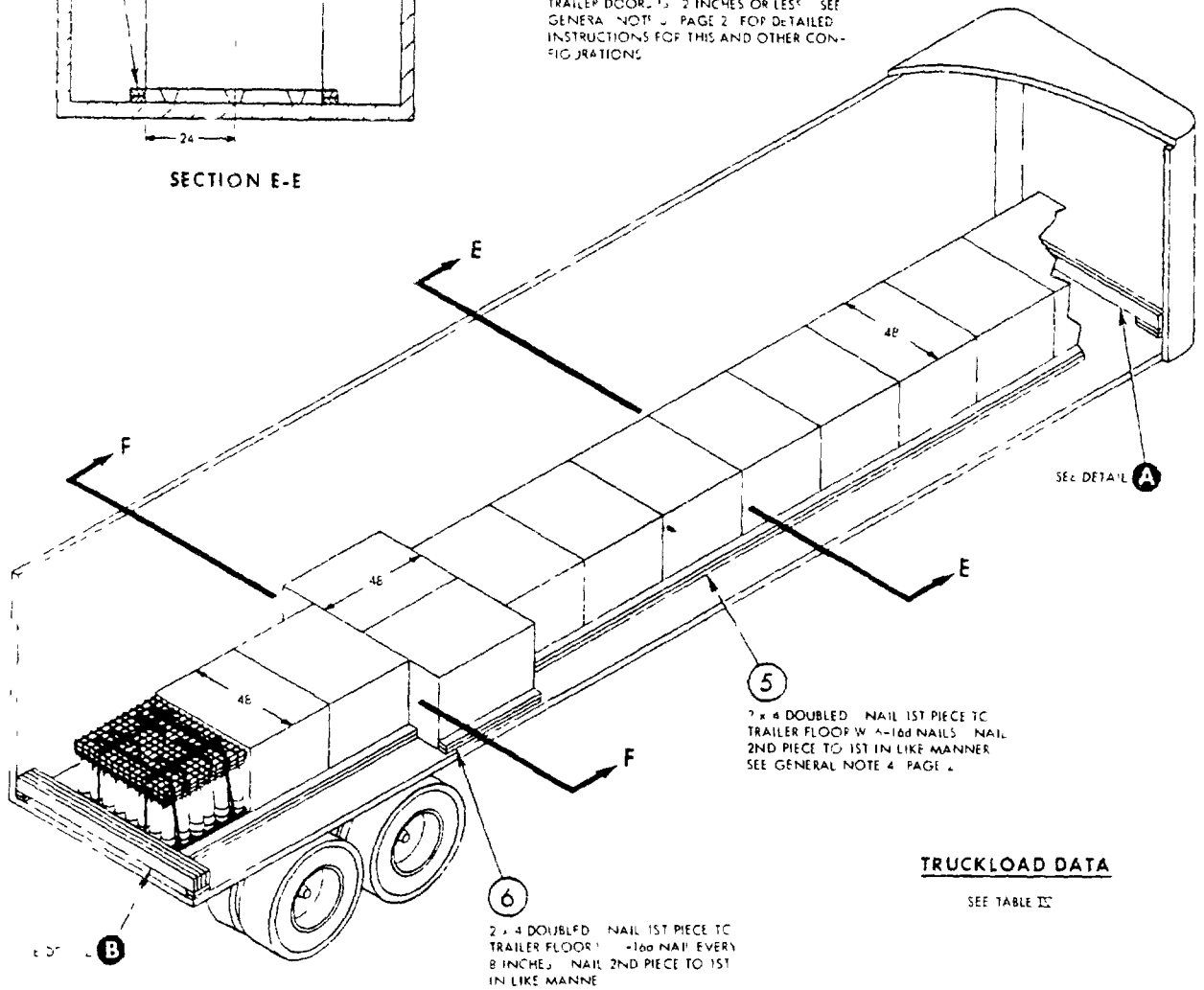
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SECTION E-E

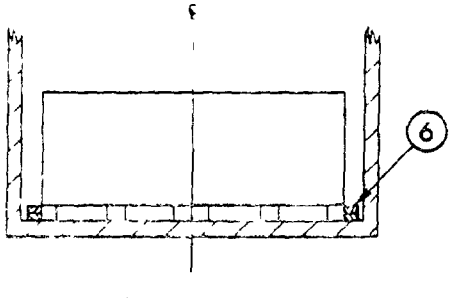
NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOOR IS 2 INCHES OR LESS SEE GENERAL NOTE 5 PAGE 2 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS



TRUCKLOAD DATA

SEE TABLE IX



SECTION F-F

*TW TRAILER WIDTH

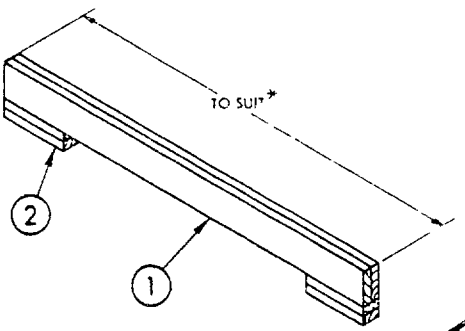
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS AND NAILING DATA				NAILS	
6	SLEEPER	2 x 4 x 48	4	SEE FIELD NOTE		
5	SLEEPER	2 x 4 x RANDOM LENGTHS	AS REQD	SEE FIELD NOTE		
4	SUPPORT	2 x 4 TO SUIT			JOINT	10d
					JOINT	10d
3	HORIZONTAL	THICKNESS TO SUIT x 6 x TW* -1	AS REQD	LAMINATE	AS REQD	10d
2	SUPPORT	2 x 4 x 12	4		JOINT	10d
					JOINT	10d
1	HORIZONTAL	2 x 6 TO SUIT	2	LAMINATE	6	10d

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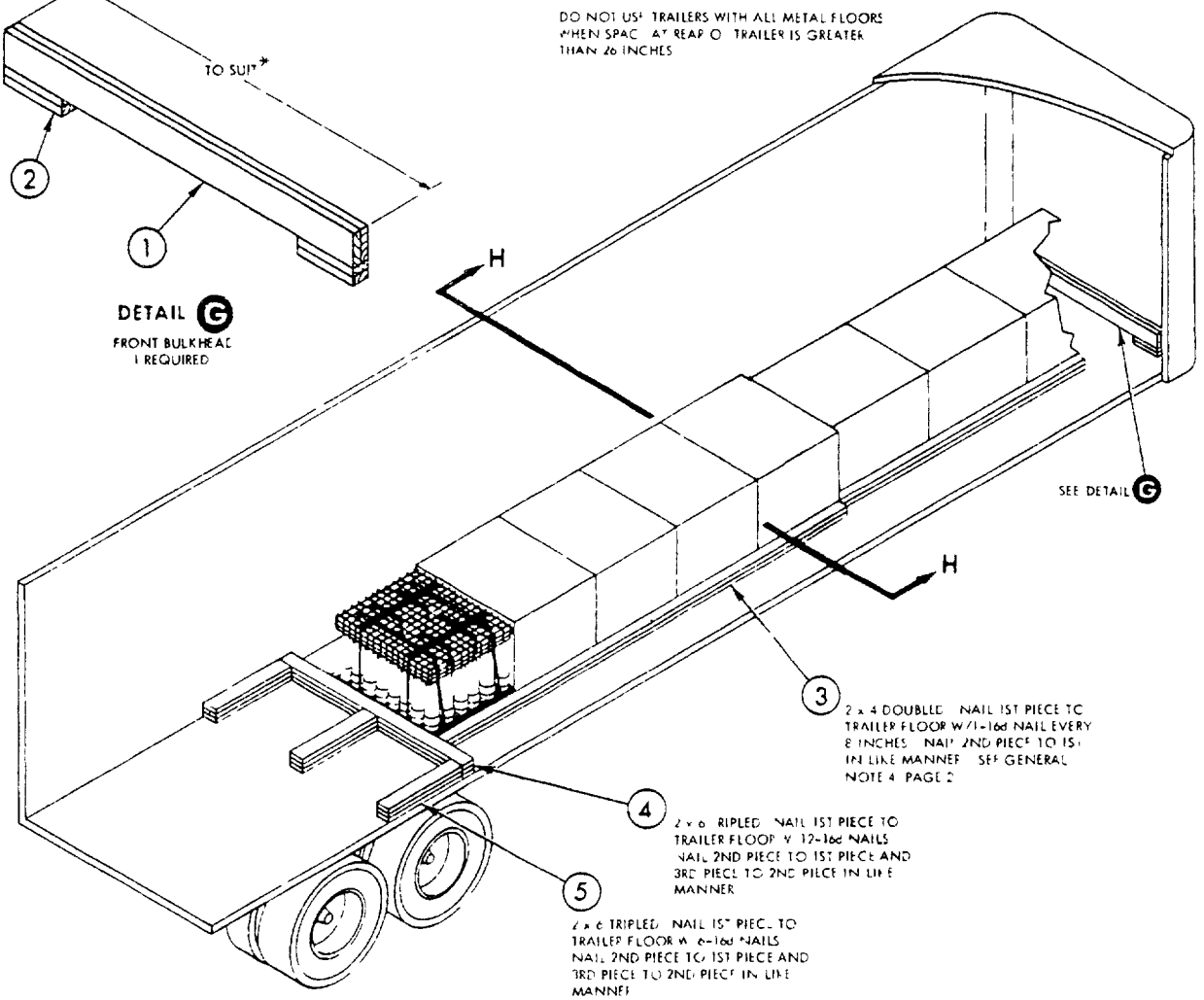
*WIDTH OF TRAILER AT FRONT WALL (DOES NOT BEAR AGAINST CORNER RADIUS

WARNING

DO NOT USE TRAILERS WITH ALL METAL FLOORS WHEN SPACE AT REAR OF TRAILER IS GREATER THAN 26 INCHES



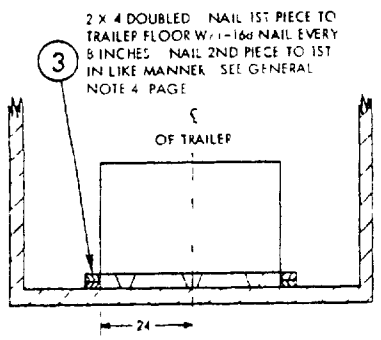
DETAIL G
FRONT BULKHEAD IF REQUIRED



3 — 2 x 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST PIECE IN LIKE MANNER SEE GENERAL NOTE 4 PAGE 2

4 — 2 x 6 RIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/ 12-16d NAILS NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER

5 — 2 x 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/ 6-16d NAILS NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER



SECTION H-H

*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
5	BACKUP CLEAT	2 x 6 x 6	6	SEE FIELD NOTE		
4	CROSSMEMBER	2 x 6 x TW* -		SEE FIELD NOTE		
3	SLEEPER	2 x 4 x RANDOM LENGTH	AS REQD	SEE FIELD NOTE		
2	SUPPORT	2 x 4 x 1	4	1	3 JOINT	10d
				2	3 JOINT	10d
1	HORIZONTAL	2 x 6 TO SUIT	2	LAMINATE	6	10d

LIST OF MATERIALS AND NAILING DATA

TYPICAL LTL

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
PROJECT NO 8140-N286

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-STD-1320-152 (Navy)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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